Enrollment No.: _____

Darshan UNIVERSITY

Darshan Institute of Computer Application

MCA | Sem-3 | Winter-2024

Course Code	:	2305CS332	**Date**	:	25-10-2024
Course Name	:	Computer Network	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:
1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

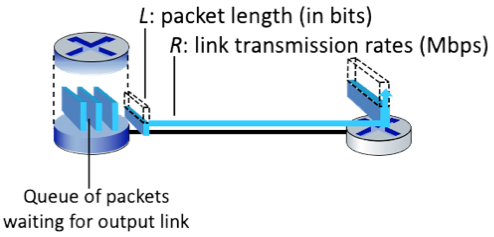
Q.1 **(A)** Define computer network with its characteristics. **4**

(B) What is Computer Network? Explain Advantages and Disadvantages of Computer Network. **3**

OR

Explain types of computer network. (LAN, MAN and WAN)

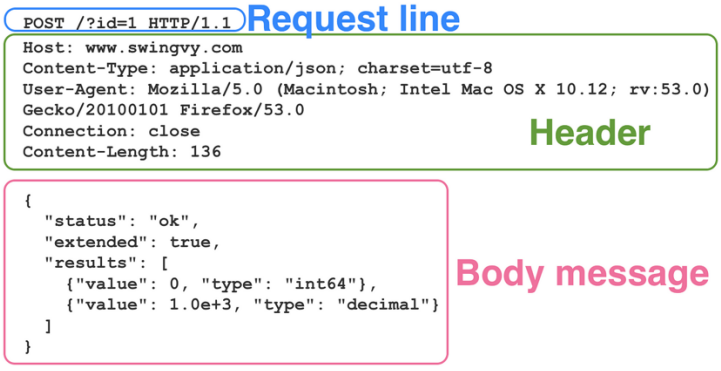
(C) Suppose a packet is L = 1500 bytes long, and link transmits at R = 1 Gbps. Derive the transmission delay for this packet? **7**



OR

Draw and explain TCP/IP protocol model.

Q.2 **(A)** Identify status code, URL, Browser name, Operating System used in HTTP message. **4**



(B) Define HTTP response status code: 404, 505, 200. **3**

OR

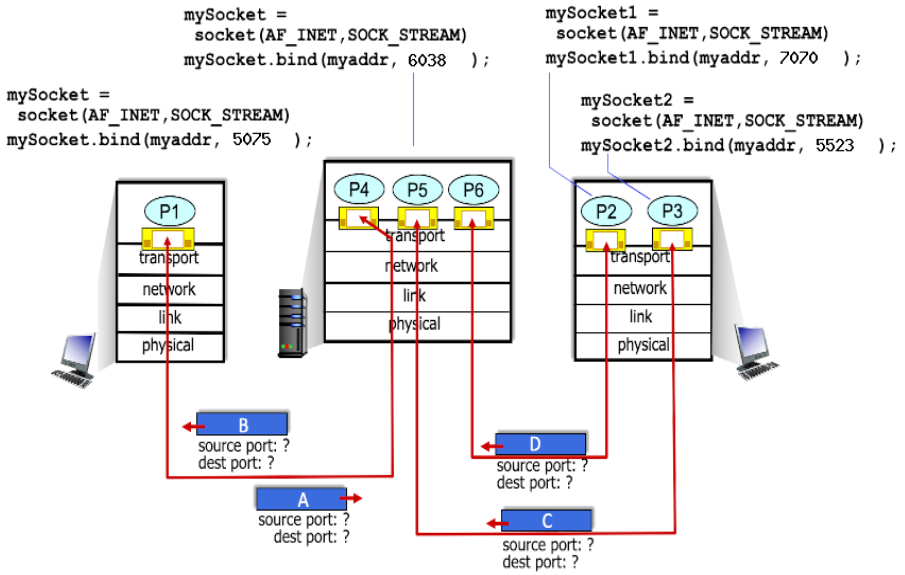
State that why distributed database design is more preferred over centralized design in DNS?

(C) Define HTTP. Explain types of HTTP with request – response behavior diagram. **7**

OR

Explain SMTP protocol with proper diagram.

Q.3 **(A)** Observe below diagram: **4**



1) What is the source port # for packet D?
2) What is the destination port # for packet D?
3) What is the source port # for packet B?
4) What is the destination port # for packet B?

(B) Draw UDP segment Header. **3**

OR

Define Go-Back-N protocol. How it works?

(C) Derive checksum value for sender side and receiver side and identify error introduced in data or not? **7**
Sender data: 1001100001101000111100000000111 (Use 8-bit word)

OR

Explain rdt 2.0 with proper diagram.

Q.4 **(A)** Given IP address 169.25.45.40/20. Carry out below details: **4**

1) Class
2) Subnet mask
3) What are the total subnets?
4) What are the total hosts?

(B) Draw IPv4 datagram format. **3**

OR

A block of addresses is granted to a small organization. We know that one of the addresses is 205.16.37.39/28. Find is the first address, last address, number of addresses in a block.

(C) Describe link state routing protocol with any example. **7**

OR

Explain different types of IP addressing classes.

Q.5 **(A)** Describe parity check technique with example. **4**

(B) Write a short note on FDMA. **3**

OR

Write a short note on TDMA.

(C) Describe CRC technique with given example. (Data: 1010000, Polynomial Eqa. X^3+1) **7**

OR

Discuss working of ARP protocol.

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